

Name of the subject	AI/ML Applications in Civil Engineering	
L-T-P	0-0-3	
Credits	2	
Name of the Department	Civil Engineering	
Status of the subject	(a) Semester : Spring (b) Level of subject : 2 nd Year UG (c) Nature of subject : Core (d) Semester to be offered : 3 rd (e) Programme in which the course is included: B.Tech. (H) in Civil Engineering	
Prerequisites	None	
Objectives	The objective of the course is to demonstrate different AI/ML methods in Civil Engineering problems with specific focus on convolution neural networks, image processing and support vector machine. In all experiments, the students will gather the data through real-life experimentation in the laboratory or in the field; followed by application of AI/ML routines for predicting outcomes.	
Names of the faculty members of the department who have the necessary expertise to teach the course	All faculty members of Civil Engineering	
Any overlap with existing subjects	None	
Recommended Text Books	1. Artificial Intelligence: A Modern Approach . Stuart Russel, Peter Norvig, Prentice Hall 2. Data Science from Scratch, Joel Grus, O'Reilly 3. Introduction to Machine Learning with Python, Andreas C. Muller & Sarah Guido, O'Reilly	
Topics to be Covered	Name of the Topic	No. of days
(1)	Structural Engineering Problems: Truss/Two-Three Hinged Arches	2
(2)	Hydraulic Engineering Problems: Open Channel Flow/Porous Media Flow	2
(3)	Geotechnical Engineering Problem: Direct Shear Test	2
(4)	Transportation Engineering Problems: Traffic Planning/Pavement Material Testing/Los Angeles Abrasion Test	2
(5)	Environmental Engineering Problem: Contaminant Diffusion in Air	2